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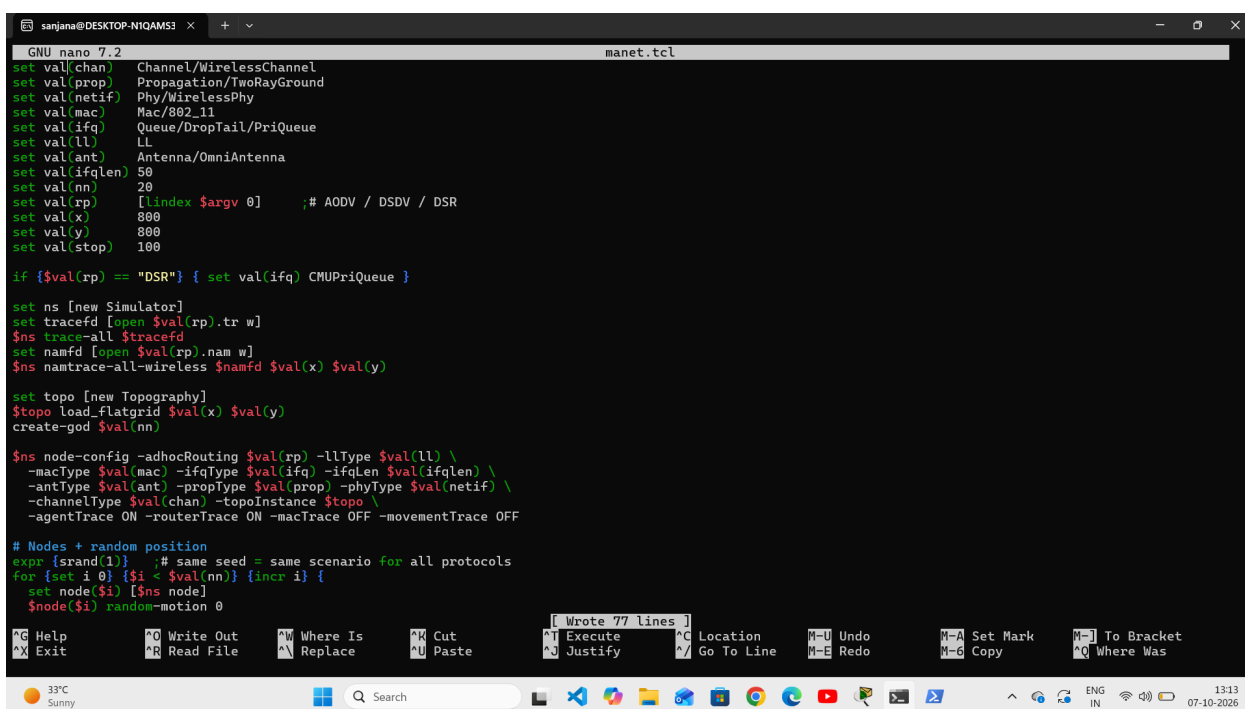
AIM: Performance Evaluation of MANET Protocols

Measure throughput, packet delivery ratio, and end-to-end delay for different MANET routing protocols.

nano manet.tcl

ns manet.tcl AODV ns manet

```
sanjana@DESKTOP-N1QAMS3:~$ mkdir ~/manet
sanjana@DESKTOP-N1QAMS3:~$ cd ~/manet
sanjana@DESKTOP-N1QAMS3:~/manet$ nano manet.tcl
sanjana@DESKTOP-N1QAMS3:~/manet$ ns manet.tcl AODV
num_nodes is set 20
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!
sanjana@DESKTOP-N1QAMS3:~/manet$ ls
AODV.nam  AODV.tr  manet.tcl
```



```
GNU nano 7.2 manet.tcl
set val(chan) Channel/WirelessChannel
set val(prop) Propagation/TwoRayGround
set val(netif) Phy/WirelessPhy
set val(mac) Mac/802_11
set val(ifq) Queue/DropTail/PriQueue
set val(ll) LL
set val(ant) Antenna/OmniAntenna
set val(ifqlen) 50
set val(nn) 20
set val(rp) [lindex $argv 0] ;# AODV / DSDV / DSR
set val(x) 800
set val(y) 800
set val(stop) 100

if {$val(rp) == "DSR"} { set val(ifq) CMUPriQueue }

set ns [new Simulator]
set tracefd [open $val(rp).tr w]
$ns trace-all $tracefd
set namfd [open $val(rp).nam w]
$ns namtrace-all-wireless $namfd $val(x) $val(y)

set topo [new Topography]
$topo load_flatgrid $val(x) $val(y)
create-god $val(nn)

$ns node-config -adhocRouting $val(rp) -llType $val(ll) \
-macType $val(mac) -ifqType $val(ifq) -ifqLen $val(ifqlen) \
-antType $val(ant) -propType $val(prop) -phyType $val(netif) \
-channelType $val(chan) -topoInstance $topo \
-agentTrace ON -routerTrace ON -macTrace OFF -movementTrace OFF

# Nodes + random position
expr {rand(1)} ;# same seed = same scenario for all protocols
for {set i 0} {$i < $val(nn)} {incr i} {
    set node($i) [$ns node]
    $node($i) random-motion 0
}

[ Wrote 77 lines ]
[ Execute Justify ]
[ Location Go To Line ]
[ M-U Undo M-E Redo ]
[ M-A Set Mark M-G Copy ]
[ M-] To Bracket M-Q Where Was ]
[ H Help W Write Out W Where Is C Cut E Execute J Justify L Location G Go To Line U Undo R Read File M-M Replace A Paste M-A Set Mark M-G Copy M-] To Bracket M-Q Where Was ]
[ X Exit R Read File W Where Is C Cut E Execute J Justify L Location G Go To Line U Undo R Read File M-M Replace A Paste M-A Set Mark M-G Copy M-] To Bracket M-Q Where Was ]
```

nano metrics.awk

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```
GNU nano 7.2 metrics.awk
BEGIN { sent=0, rcv=0, bytes=0, delay=0, start=1e9, stop=0 }
if ($1=="s" && $4=="AGT" && $7=="cbr") {
    sent++; st[$6]=$2
    if ($2>start) start=$2
}
if ($1=="r" && $4=="AGT" && $7=="cbr") {
    rcv++; bytes+=$8, delay+=$(2-st[$6])
    if ($2>stop) stop=$2
}
END {
    printf "Packets Sent      : %d\n", sent
    printf "Packets Received   : %d\n", rcv
    printf "PDR (%)             : %.2f\n", (rcv/sent)*100
    printf "Throughput (kbps)   : %.2f\n", (bytes*8)/(stop-start)/1000
    printf "Avg E2E Delay (ms) : %.2f\n", (delay/rcv)*1000
}
```

awk -f metrics.awk AODV.tr

```
sanjana@DESKTOP-N1QAMS3:~/manet$ nano metrics.awk
sanjana@DESKTOP-N1QAMS3:~/manet$ awk -f metrics.awk AODV.tr
Packets Sent      : 4505
Packets Received   : 4239
PDR (%)           : 94.10
Throughput (kbps) : 200.32
Avg E2E Delay (ms): 156.45
```

awk -f metrics.awk DSDV.tr

```
sanjana@DESKTOP-N1QAMS3:~/manet$ ns manet.tcl DSDV
num_nodes is set 20
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!

sanjana@DESKTOP-N1QAMS3:~/manet$ awk -f metrics.awk DSDV.tr
Packets Sent      : 4505
Packets Received   : 2186
PDR (%)           : 48.52
Throughput (kbps) : 103.34
Avg E2E Delay (ms): 32.35
```

awk -f metrics.awk DSR.tr

```
sanjana@DESKTOP-N1QAMS3:~/manet$ ns manet.tcl DSR
num_nodes is set 20
warning: Please use -channel as shown in tcl/ex/wireless-mitf.tcl
INITIALIZE THE LIST xListHead
channel.cc:sendUp - Calc highestAntennaZ_ and distCST_
highestAntennaZ_ = 1.5, distCST_ = 550.0
SORTING LISTS ...DONE!

sanjana@DESKTOP-N1QAMS3:~/manet$ awk -f metrics.awk DSR.tr
Packets Sent      : 4505
Packets Received   : 4469
PDR (%)           : 99.20
Throughput (kbps) : 203.30
Avg E2E Delay (ms): 159.53
```

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